

Reference Benchmarks for the CoQui PAW + ISDF–THC Validation Suite: Band Gaps and RPA Energies

Compiled from the published literature

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Abstract

This document is a reference compendium for the CoQui PAW + ISDF–THC validation suite. It collects published quasiparticle band gaps (G_0W_0 , GW_0 , scGW) and RPA total/correlation energies for benchmark solids, spanning VASP, ABINIT, BerkeleyGW, GPAW, FHI–aims, exciting, SPEX/FLEUR and Quantum ESPRESSO. Two systems anchor the validation: Si (the standard “easy” benchmark) and ZnO (the well-known “hard” benchmark). Secondary systems — diamond, MgO, LiF, NaCl, GaAs, hBN, graphite, noble-gas solids, and selected metals — are included for completeness. For each entry we record the level of theory, code, projector type (PAW / ONCV / all-electron LAPW / NAO), k-mesh, band cutoff, frequency treatment, and the source DOI. The document also flags published convergence pitfalls (notably the ZnO “false-convergence trap” and the basis-set $E_{\text{cut}}^{-3/2}$ tail in RPA) so they can be avoided when comparing CoQui results to the literature.

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1 Executive summary

1.1 What to target for CoQui

Table 1: Headline targets for the first round of CoQui PAW+ISDF+THC validation. Numbers are typical converged values from the literature; see the body of the document for full tables and sources.

System	Quantity	Target value	Reference	Difficulty
Si	G_0W_0 @PBE indirect gap	$\approx 1.1\text{--}1.2\text{ eV}$	multi-code 2024	easy
Si	RPA@PBE E_{coh}	$\approx 4.6\text{ eV/atom}$	Harl 2010 (VASP)	easy
Si	RPA@PBE lattice constant	$5.421\text{ \AA}$	Harl 2010 (VASP)	easy
ZnO	G_0W_0 @PBE direct gap	$2.5\text{--}2.9\text{ eV}$	multi-code 2024	hard
ZnO	RPA@PBE lattice a	$\approx 3.254\text{ \AA}$	Harl 2010 (VASP)	hard

1.2 Three things to watch

1. **Always extrapolate $E_{\text{cut}}^{-3/2}$ for RPA.** Single-cutoff RPA correlation energies are not comparable across codes; the tail is large (0.3–0.5 eV/atom for Si at typical cutoffs).
2. **ZnO needs Zn 3s/3p/3d in valence and large virtual space** for both GW and RPA. Anything less than 1000 bands plus extrapolation is the well-documented false-convergence trap.
3. **Mind the starting point.** RPA@LDA and RPA@PBE differ by $\sim 0.2\text{--}0.3\text{ eV/atom}$ for covalent solids; G_0W_0 @LDA and G_0W_0 @PBE/PBEsol differ by $\sim 0.1\text{ eV}$ for Si. Compare apples-to-apples.

1.3 Bottom-line accuracy expectations

For Si, the four-code 2024 benchmark (Borlido et al., arXiv:2411.19701) gives a 112 meV cross-code spread for the G_0W_0 @PBEsol gap. For ZnO the same benchmark shows 334 meV spread — the largest of any solid in the test set. Anything CoQui produces inside the 2.5–2.9 eV window for ZnO with proper convergence is consistent with the established literature; values below $\sim 2.0\text{ eV}$ indicate a PAW augmentation or basis-set issue.

2 Part I — GW band gaps

2.1 Silicon (Si, diamond structure)

Structure: Fd-3m, $a = 5.431\text{\AA}$. The gap is indirect $\Gamma \rightarrow \Delta$ (about $0.85 \times X$). Experimental indirect gap: **1.17 eV** (0 K, no zero-point renormalization).

Table 2: G_0W_0 , GW_0 , scGW gap of Si.

Level	Gap (eV)	Code	PP	k-mesh	Freq.	Source
LDA (KS)	0.51	VASP	PAW	—	—	textbook
G_0W_0 @LDA	0.85	VASP	PAW	$4 \times 4 \times 4$	PP (GN)	Shishkin/Kresse 2007
GW_0 @LDA	1.22	VASP	PAW	$4 \times 4 \times 4$	PP	Shishkin/Kresse 2007
scGW@LDA	1.41	VASP	PAW	$4 \times 4 \times 4$	PP	Shishkin/Kresse 2007
G_0W_0 @LDA	1.094	VASP	PAW	$10 \times 10 \times 10$	CD	VASP wiki tutorial
G_0W_0 @PBEsol	1.257	ABINIT	ONCV	$8 \times 8 \times 8$	CD+AC	Borlido 2024
G_0W_0 @PBEsol	1.145	exciting	LAPW+lo	$8 \times 8 \times 8$	32 imag. freq.	Borlido 2024
G_0W_0 @PBEsol	1.147	FHI-aims	NAO (all-e)	$8 \times 8 \times 8$	60 freq. (Padé)	Borlido 2024
G_0W_0 @PBEsol	1.152	GPAW	PW (NC-PP)	$8 \times 8 \times 8$	real-freq. extrap.	Borlido 2024
Experiment	1.17	—	—	—	—	standard

Highlights. G_0W_0 @LDA underestimates by $\sim 0.3\text{ eV}$. Eigenvalue self-consistency in G (GW_0) closes the gap to within 0.05 eV of experiment. The 2024 multi-code benchmark with PBEsol orbitals gives $1.145\text{--}1.257\text{ eV}$; all-electron codes (FHI-aims, exciting) cluster at $\sim 1.15\text{ eV}$. The ABINIT outlier at 1.257 eV reflects residual ONCV / ϵ_{cut} sensitivity.

2.2 ZnO (wurtzite)

Structure: $P6_3mc$, $a = 3.250\text{\AA}$, $c = 5.207\text{\AA}$. The gap is *direct* $\Gamma \rightarrow \Gamma$. Experimental optical gap: **3.44 eV** (4 K). Zn 3d band at $\sim -7\text{ eV}$. LDA gives $\sim 0.7\text{--}0.8\text{ eV}$.

Table 3: G_0W_0 gap of ZnO — the canonical hard benchmark.

Level	Gap (eV)	Code	PP	Zn 3d	k-mesh	Freq.	Source
LDA (KS)	0.76	ABINIT	ONCV	val.	$8 \times 8 \times 5$	—	Borlido 2024
G_0W_0 @LDA	~ 0.8	VASP	PAW	core	$4 \times 4 \times 4$	PP	Shishkin/Kresse 2007
GW_0 @LDA	~ 2.97	VASP	PAW	core	$4 \times 4 \times 4$	PP	Shishkin/Kresse 2007
G_0W_0 @LDA conv.	~ 2.5	BerkeleyGW	NC-PP	val.	$4 \times 4 \times 3$	GPP	Shih 2010
G_0W_0 @LDA conv.	$3.0\text{--}3.2$	SPEX	FLAPW	val.	$4 \times 4 \times 4$	full-freq.	Friedrich 2011
G_0W_0 @LDA (CD)	~ 2.4	ABINIT	PAW	val.	$6 \times 6 \times 4$	CD	Stankovski 2011
G_0W_0 @LDA+U	~ 3.2	VASP	PAW	val.	$6 \times 6 \times 4$	PP	Lany/Zunger 2010

Table 3 continued.

Level	Gap (eV)	Code	PP	Zn 3d	k-mesh	Freq.	Source
G_0W_0 @PBEsol	2.613	ABINIT	ONCV	val.	$8 \times 8 \times 5$	CD+AC	Borlido 2024
G_0W_0 @PBEsol	2.878	exciting	LAPW	val.	$8 \times 8 \times 5$	32 imag.	Borlido 2024
G_0W_0 @PBEsol	2.758	FHI-aims	NAO	val.	$8 \times 8 \times 5$	60 freq.	Borlido 2024
G_0W_0 @PBEsol	2.544	GPAW	PW	val.	$8 \times 8 \times 5$	real-freq.	Borlido 2024
Experiment (optical)	3.44	—	—	—	—	—	standard
Experiment (transport)	3.37	—	—	—	—	—	standard

The convergence saga. The published ZnO G_0W_0 value drifted from ~ 0.8 eV (2007) to ~ 2.5 – 2.9 eV (2024) as two issues were resolved. (1) Zn 3s/3p/3d must be in valence, not core. (2) The number of empty bands must be $\gg 1000$ with explicit $E_{\text{cut}}^{-3/2}$ extrapolation, because the Coulomb-hole self-energy Σ_{COH} shows *non-uniform convergence*: VBM and CBM converge at different rates, biasing the gap. f-sum-rule plasmon-pole models are unreliable for ZnO because the shallow Zn 3d resonance pollutes the f-sum; contour deformation or full-frequency integration is required.

2.3 Secondary GW benchmarks

MgO (rock-salt, Fm-3m). Direct $\Gamma \rightarrow \Gamma$ gap; experiment **7.83 eV**; LDA ~ 4.5 eV. VASP PAW (Shishkin/Kresse 2007): G_0W_0 @LDA ~ 6.2 – 6.5 eV, GW_0 @LDA ~ 7.2 – 7.5 eV, scGW@LDA ~ 7.8 eV.

LiF (rock-salt, Fm-3m). Direct gap; experiment **14.2 eV**; LDA ~ 9.2 eV. VASP PAW: G_0W_0 @LDA 12.0–12.5 eV, GW_0 @LDA ~ 13.5 eV, scGW@LDA ~ 14.8 eV (overshoots).

GaAs (zinc-blende, F-43m). Direct gap, experiment **1.52 eV** at 0 K; LDA ~ 0.2 – 0.5 eV; Ga 3d at ~ -18 eV. VASP PAW (Ga 3d as core): G_0W_0 @LDA ~ 1.1 – 1.3 eV, GW_0 @LDA ~ 1.5 eV. BerkeleyGW (Ga 3d valence): G_0W_0 @LDA 1.3–1.5 eV.

TiO₂ rutile (P4₂/mnm). Direct optical gap, experiment **3.03 eV**; LDA ~ 1.8 eV. 2024 cross-code G_0W_0 @PBEsol: ABINIT 3.333, exciting 3.321, FHI-aims 3.546, GPAW 3.281 eV.

2.4 Cross-code precision benchmark (2024)

Source: Borlido et al., *arXiv:2411.19701* / Comput. Mater. Sci. 2025, DOI 10.1016/j.commatsci.2024.113495. PBEsol orbitals; full-frequency methods (no plasmon-pole). Zn valence in ABINIT ONCV is 20 electrons; exciting and FHI-aims are all-electron.

Table 4: Four-code G_0W_0 @PBEsol gaps for Si, ZnO, TiO₂ (eV). Spread is the max–min range across the four codes.

Material	ABINIT	exciting	FHI-aims	GPAW	Spread (meV)	Expt
Si (indirect)	1.257	1.145	1.147	1.152	112	1.17
ZnO (direct)	2.613	2.878	2.758	2.544	334	3.44
TiO ₂ rutile (dir.)	3.333	3.321	3.546	3.281	265	3.03

2.5 Six-step ZnO convergence checklist

1. **Semicore in valence.** Zn 3s, 3p, 3d in the active space (12 vs 20 valence electrons). Core treatment $\Rightarrow \sim 0.8$ eV gap, wrong by ~ 2.6 eV.
2. **Empty bands.** ~ 1000 – 3000 bands or basis-set extrapolation; Σ_{COH} converges non-uniformly.
3. **χ cutoff.** $\text{ENCUTGW} \gtrsim 30$ – 50 Ha in plane-wave codes; otherwise a false convergence plateau.
4. **Frequency scheme.** Avoid f-sum-rule plasmon-pole; use contour deformation or full-frequency integration.
5. **Starting point.** LDA places Zn 3d too high (-5 vs -7 eV). LDA+U or hybrid starting points improve convergence.
6. **PAW partial-wave completeness.** Standard PAW partial waves are incomplete for high-energy unoccupied states; norm-conserving partial waves or basis-set corrections (Klimeš/Kaltak/Kresse, PRB 90, 075125, 2014) are needed for predictive results.

3 Part II — RPA total / correlation energies

Convention. In what follows “RPA” means RPA correlation on top of a DFT exchange piece, i.e. $E_{\text{tot}} = E_x[\phi^{\text{KS}}] + E_c^{\text{RPA}}[\chi_0]$. “RPA@PBE” is the standard choice in the literature (Harl/Kresse). Cohesive energies are reported as positive = bound, in eV/atom unless noted.

3.1 Silicon

Experimental cohesive energy: **4.62 eV/atom** (0 K, ZPE-corrected). Lattice constant: **5.431 Å**; bulk modulus **99 GPa**.

Table 5: Si cohesive energy at the RPA level.

Method	E_{coh} (eV/atom)	Code	PP	k-mesh	Source
PBE	4.55	VASP	PAW	$8 \times 8 \times 8$	Harl 2010
RPA@PBE	~ 4.6	VASP	PAW	$8 \times 8 \times 8$	Harl 2010
RPA@LDA	4.32	GPAW	PAW	$4 \times 4 \times 4$	Olsen/Thygesen 2013
EXX@PBE (no corr.)	2.82	GPAW	PAW	$4 \times 4 \times 4$	GPAW tutorial
Experiment	4.62	—	—	—	standard

Table 6: Si lattice constant and bulk modulus, RPA vs PBE vs experiment.

Method	a (Å)	B (GPa)	Code	Source
PBE	5.468	90	VASP/PAW	Harl 2010
RPA@PBE	5.421	~ 98	VASP/PAW	Harl 2010
RPA	5.414	—	QE/ONCV	Pitts 2025
PBE	5.479	—	QE/ONCV	Pitts 2025
Expt.	5.431	99	—	standard

3.2 Diamond, MgO, LiF, NaCl

Compact summary; full per-system tables in `notes/paw_rpa_reference_benchmarks.md`.

Table 7: RPA@PBE structural and cohesive properties (VASP/PAW, Harl 2010, unless noted). Experimental values in the right column.

Material	a_{PBE} (Å)	a_{RPA} (Å)	a_{expt} (Å)	$E_{\text{coh}}^{\text{RPA@PBE}}$ (eV/at.)	$E_{\text{coh}}^{\text{expt}}$ (eV/at.)	Notes
Si	5.468	5.421	5.431	~ 4.6	4.62	covalent
C (diamond)	3.572	3.553	3.553	$\sim 7.4\text{--}7.5$	7.37	covalent, $B = 443$ GPa
BN (zinc-blende)	3.626	3.604	3.592	—	—	—
MgO	4.258	4.196	4.211	$\sim 5.0\text{--}5.1$	5.20	ionic
LiF	4.073	3.986	3.972	$\sim 4.3\text{--}4.4$	4.46	ionic
NaCl	5.694	5.573	5.595	~ 3.3	3.34	soft ionic
ZnO (a)	3.283	3.254	3.250	—	$\sim 3.60/\text{atom}$	semicore
ZnO (c)	5.304	5.262	5.207	—	—	semicore

3.3 Layered and weakly-bound systems

Graphite interlayer binding. The landmark RPA-for-graphite paper (Lebègue et al., PRL 105, 196401, 2010): VASP/PAW RPA@PBE gives an interlayer binding energy of **48 meV/atom** at $d_{\text{int}} = 3.34 \text{ \AA}$, in agreement with the experimental window 35–52 meV/atom. PBE gives essentially zero interlayer binding. RPA captures the correct $1/d^3$ van der Waals asymptotics.

Hexagonal BN. RPA@PBE in-plane $a \approx 2.50 \text{ \AA}$, interlayer binding 30–40 meV/atom (VASP/PAW, Harl 2010). PBE: no binding.

Noble-gas solids. Harl & Kresse, PRB 77, 045136 (2008), VASP/PAW, RPA@LDA. The foundational RPA-for-vdW-solids paper.

Table 8: Noble-gas solids: lattice constant and cohesive energy at RPA@LDA vs experiment.

System	a_{RPA} (Å)	a_{expt} (Å)	$E_{\text{coh}}^{\text{RPA}}$ (meV/at)	$E_{\text{coh}}^{\text{expt}}$ (meV/at)
Ne	~ 4.46	4.430	~ 27	27
Ar	~ 5.26	5.256	~ 88	88
Kr	~ 5.64	5.646	~ 120	116

3.4 Metals

Schimka et al., *Nature Materials* 9, 741 (2010) and PRB 87, 214102 (2013). RPA@PBE cohesive energies for bulk metals are roughly PBE-quality; the gain from RPA is concentrated in surface and adsorption energies, not bulk cohesion. Approximate values (eV/atom): Cu ~ 3.5 , Fe ~ 4.3 , Na ~ 1.1 , Mg ~ 1.5 , Al ~ 3.4 .

3.5 Statistical summary across 20 solids (Harl 2010)

- PBE lattice constant MAE $\approx 0.05 \text{ \AA}$ (systematic overestimate); RPA@PBE lattice constant MAE $\approx 0.02 \text{ \AA}$ (mixed).
- PBE bulk modulus MAE $\approx 9 \text{ GPa}$; RPA@PBE MAE $\approx 5 \text{ GPa}$.
- RPA@PBE is a clear improvement over PBE for structural properties.
- RPA *underbinds* covalent solids (Si, C) by 0.1–0.3 eV/atom relative to experiment; ionic solids fare better.

3.6 Convergence parameters by code

The $E_{\text{cut}}^{-3/2}$ tail. The leading correction to a finite-cutoff RPA correlation energy is $E_{\text{RPA}}(E_{\text{cut}}) \approx E_{\text{RPA}}(\infty) + A E_{\text{cut}}^{-3/2}$ (Harl/Kresse 2008). For Si at ENCUT $\sim 400 \text{ eV}$ the missing tail is 0.3–0.5 eV/atom. *Always extrapolate.* Single-cutoff numbers are not portable across codes.

Table 9: Recommended RPA convergence parameters by code.

Code	Cutoff for χ_0	N_ω	N_{bands}	Extrapolation
VASP/PAW	ENCUTGW = ENCUT (or 2/3)	12–16 Gauss- Legendre	$\sim 3\text{--}5 \times N_{\text{occ}}$	$E_{\text{cut}}^{-3/2}$ on E_{RPA}
GPAW/PAW	80–164 eV (χ_0 cutoff)	16 GL	set by N_{pw}	$E_{\text{cut}}^{-3/2}$
FHI-aims	all-electron NAO tier-2/3	40–60 freq.	imag. all virtual NAO	NAO basis tier
QE/ONCV	~ 80 Ry	8 imag. freq.	$\sim 5 \times N_e$	basis ext. implicit

4 Recommended validation order for CoQui

1. **Si** — both G_0W_0 and RPA. The simplest check on the full PAW + ISDF-THC pipeline. Targets: G_0W_0 @PBE gap ≈ 1.1 eV, RPA@PBE $E_{\text{coh}} \approx 4.6$ eV/atom, $a_{\text{RPA}} \approx 5.421 \text{ \AA}$.
2. **MgO and LiF** — ionic solids where RPA performs well. Tests Hartree/exchange balance in the PAW augmentation. Targets: $a_{\text{MgO}} \approx 4.196\text{--}4.200 \text{ \AA}$, $a_{\text{LiF}} \approx 3.986\text{--}3.996 \text{ \AA}$.
3. **Diamond** — covalent, small cell, well-studied. Target $a \approx 3.553 \text{ \AA}$, RPA@PBE $E_{\text{coh}} \approx 7.4\text{--}7.5$ eV/atom.
4. **ZnO** — semicore-state stress test. Zn 3s/3p/3d in valence, ≥ 1000 bands with extrapolation, contour deformation. Targets: G_0W_0 @PBE in 2.5–2.9 eV window; $a_{\text{RPA}} \approx 3.254 \text{ \AA}$. Anything below ~ 2.0 eV for the ZnO gap indicates a PAW augmentation or basis-set issue.

5 Key references

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6. Borlido et al., arXiv:2411.19701 / *Comput. Mater. Sci.* 2025. DOI 10.1016/j.commatsci.2024.113495. Four-code G_0W_0 precision benchmark (ABINIT, exciting, FHI-aims, GPAW); Si, ZnO, TiO₂. **Most current multi-code reference.**

7. Harl & Kresse, *Phys. Rev. B* **77**, 045136 (2008). DOI 10.1103/PhysRevB.77.045136. Foundational RPA-for-solids paper: VASP/PAW ACFDT for Ne, Ar, Kr; establishes $E_{\text{cut}}^{-3/2}$ basis extrapolation.
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16. Schimka et al., *Phys. Rev. B* **87**, 214102 (2013). DOI 10.1103/PhysRevB.87.214102. RPA@PBE lattice constants and cohesive energies of alkali / alkaline-earth / transition metals.

6 Provenance and uncertainty

The numerical values in this document are drawn from the cited primary references where the published tables are accessible, and from review abstracts / cross-citations in newer papers where the original tables are paywalled (notably Harl 2010 PRB 81, 115126). Values quoted with a “~” carry approximately ± 0.1 eV (or ± 0.01 Å for lattice constants) of indirect-sourcing uncertainty. The DOIs are included so the original tables can be consulted when institutional access is available. The companion markdown files (`notes/paw_gw_reference_benchmarks.md` and `notes/paw_rpa_reference_benchmarks.md`) include longer annotations and per-row sources.